

MEGH-ASHISH SHAH
CSE - A

ASSIGNMENT - 1

SUBJECT:- EMT
TOPIC: SIMULTANEOUS
EQUATIONS

DATE: _____
ROLL NO.: A051

EXAMPLE :-

$$4x + 7y = 5 \quad \text{--- (1)} \quad 3x + 9y = 10 \quad \text{--- (2)}$$

Find x and y

By Normal Method

Multiply eq. (1) by 3 and eq. (2) with 4
and subtract

$$\begin{array}{rcl} 12x + 21y & = & 15 \\ (-) 12x + 36y & = & 40 \\ \hline & & (-) \quad (-) \quad (-) \\ & & -15y = -25 \end{array}$$

$$-15y = -25$$

$$15y = 25$$

$$\boxed{y = \frac{5}{3}}$$

Substituting $y = \frac{5}{3}$ in (1)

$$4x + 7 \times \frac{5}{3} = 5$$

$$4x + \frac{35}{3} = 5$$

$$4x = 5 - \frac{35}{3}$$

$$4x = \frac{15 - 35}{3} = -\frac{20}{3}$$

$$\therefore x = -\frac{20}{3} \times \frac{1}{4} \quad \therefore \boxed{x = -\frac{5}{3}}$$

Now, we will be using techniques provided by Vedic Math to solve simultaneous equations. It can be divided in 3 categories:-

Category 1:- When the ratio of coefficients of either x or y is the same as that of the constants.

Example :-

$$3x + 4y = 10$$

$$5x + 8y = 20$$

Since ratio of co-efficients of y is $4/8 = 1/2$ and ratio of constants is $10/20 = 1/2$, in such equations value of x is 0.

Therefore we can substitute $x = 0$

$$4y = 10$$

$$\therefore y = 5/2$$

Category 2 :- When the co-efficients of x and y are interchanged

$$\text{Example :- } 45x - 23y = 113 \quad (1)$$

$$23x - 45y = 91 \quad (2)$$

We will add and then subtract the given equation to get a simplified set as :-

$$68x - 68y = 204 \quad (3)$$

$$22x + 22y = 22 \quad (4)$$

Dividing eq(3) by 68 and eq(4) by 22

$$\therefore \begin{aligned} x - y &= 3 \quad (5) \\ x + y &= 1 \quad (6) \end{aligned}$$

This gives value of $x = 2$ and $y = -1$

Category 3 : All other types not falling in these two categories

Example :-

$$4x + 7y = 5 \quad \dots (1)$$

$$3x + 9y = 10 \quad \dots (2)$$

Let's represent this as a set of coefficients as follow :- $a - b - c$,
 $p - q - r$

Where $a = 4$, $b = 7$, $c = 5$, $p = 3$, $q = 9$, $r = 10$

We can get value of x by using the formula :-

$$x = \frac{b \cdot r - c \cdot q}{b \cdot p - a \cdot q}$$

$$\therefore x = \frac{7 \times 10 - 5 \times 9}{7 \times 3 - 4 \times 9}$$

$$\therefore x = \frac{70 - 45}{21 - 36} = -\frac{25}{15} = \boxed{-\frac{5}{3}}$$

We can get value of y by using the formula :-

$$y = \frac{c \cdot p - a \cdot q}{b \cdot p - a \cdot q}$$

$$\therefore y = \frac{5 \times 3 - 4 \times 10}{7 \times 3 - 4 \times 9}$$

$$\therefore y = \frac{15 - 40}{21 - 36} = \frac{-25}{-15} = \boxed{\frac{5}{3}}$$

EXERCISE QUESTIONS

a) $2x + 3y = 8$
 $4x + 5y = 14$

Let From Category 3

let

$$a=2, b=3, c=8, p=4, q=5, r=14$$

$$x = \frac{b \cdot r - c \cdot q}{b \cdot p - a \cdot q}$$

$$y = \frac{c \cdot p - a \cdot r}{b \cdot p - a \cdot q}$$

$$x = \frac{3 \times 14 - 8 \times 5}{3 \times 4 - 2 \times 5}$$

$$y = \frac{8 \times 4 - 2 \times 14}{3 \times 4 - 2 \times 5}$$

$$x = \frac{42 - 40}{12 - 10}$$

$$y = \frac{32 - 28}{12 - 10}$$

$$x = \frac{2}{2}$$

$$y = \frac{4}{2}$$

$$\boxed{x = 1}$$

$$\boxed{y = 2}$$

$$b) \quad \begin{aligned} x - y &= 7 \\ 5x + 2y &= 42 \end{aligned}$$

n:- let

$$a = 1, \quad b = -1, \quad c = 7, \quad p = 5, \quad q = 2, \quad r = 42$$

$$x = \frac{b \cdot r - c \cdot q}{b \cdot p - a \cdot q}$$

$$y = \frac{c \cdot p - a \cdot r}{b \cdot p - a \cdot q}$$

$$x = \frac{-1 \times 42 - 7 \times 2}{-1 \times 5 - 1 \times 2}$$

$$y = \frac{7 \times 5 - 1 \times 42}{-1 \times 5 - 1 \times 2}$$

$$x = \frac{-42 - 14}{-5 - 2}$$

$$y = \frac{35 - 42}{-5 - 2}$$

$$x = \frac{-56}{-7}$$

$$y = \frac{-7}{-7}$$

$$\boxed{x = 8}$$

$$\boxed{y = 1}$$

$$c) \quad \begin{aligned} 2x + y &= 5 \\ 3x - 4y &= 2 \end{aligned}$$

n:- let

$$a = 2, \quad b = 1, \quad c = 5, \quad p = 3, \quad q = -4, \quad r = 2$$

$$x = \frac{b \cdot r - c \cdot q}{b \cdot p - a \cdot q}$$

$$= \frac{1 \times 2 - 5 \times -4}{1 \times 3 - 2 \times -4}$$

$$= \frac{2 + 20}{3 + 8} = \frac{22}{11} = \boxed{2}$$

$$y = \frac{c^* p - a^* q}{b^* p - a^* q} = \frac{5 \times 3 - 2^* \times 2}{1 \times 3 - 2 \times -4}$$

$$= \frac{15 - 4}{3 + 8} = \frac{11}{11} = \boxed{1}$$

$$\therefore x = 2, y = 1$$

d) $5x - 3y = 11$
 $6x - 5y = 9$

Solⁿ: - Let

$$a = 5, b = -3, c = 11, p = 6, q = -5, r = 9$$

$$x = \frac{b^* r - c^* q}{b^* p - a^* q}$$

$$y = \frac{c^* p - a^* r}{b^* p - a^* q}$$

$$x = \frac{-3 \times 9 - 11 \times -5}{-3 \times 6 - 5 \times -5}$$

$$y = \frac{11 \times 6 - 5 \times 9}{-3 \times 6 - 5 \times -5}$$

$$x = \frac{-27 + 55}{-18 + 25}$$

$$y = \frac{66 - 45}{-18 + 25}$$

$$x = \frac{28}{7}$$

$$y = \frac{21}{7}$$

$$\boxed{x = 4}$$

$$\boxed{y = 3}$$

$$\begin{aligned} 11x + 6y &= 28 \\ 7x - 4y &= 10 \end{aligned}$$

1st :- let

$$a = 11, b = 6, c = 28, p = 7, q = -4, r = 10$$

$$x = \frac{b \cdot r - c \cdot q}{b \cdot p - a \cdot q}$$

$$y = \frac{c \cdot p - a \cdot r}{b \cdot p - a \cdot q}$$

$$x = \frac{6 \times 10 - 28 \times -4}{6 \times 7 - 11 \times -4}$$

$$y = \frac{28 \times 7 - 11 \times 10}{6 \times 7 - 11 \times -4}$$

$$x = \frac{60 + 112}{42 + 44}$$

$$y = \frac{196 - 110}{42 + 44}$$

$$x = \frac{172}{86}$$

$$y = \frac{86}{86}$$

$$\boxed{x = 2}$$

$$\boxed{y = 1}$$

$$\begin{aligned} 12x + 8y &= 7 \quad - (1) \\ 16x + 16y &= 14 \quad - (2) \end{aligned}$$

2nd :- Since ratio of coefficients of y
 $= 8/16 = 1/2$
 and ratio of constants $= 7/14 = 1/2$
 is same

$\therefore$ Therefore the value of x is zero and we can substitute $x = 0$ in (1)

$$\therefore y = 7/8$$

$$\therefore x = 0, y = \frac{7}{8}$$

$$\begin{array}{rcl} \text{g)} & 6x + 7y = 8 & - (1) \\ & 19x + 14y = 16 & - (2) \end{array}$$

Solⁿ: Since the ratio of coefficient of $y = 7/14 = 1/2$ and ratio of constants $= 8/16 = 1/2$ is equal

therefore x can be assumed to be zero and we can substitute $x = 0$ in (1)

$$\therefore 7y = 8$$

$$y = \frac{8}{7}$$

$$\therefore x = 0, y = \frac{8}{7}$$

$$\begin{array}{rcl} \text{h)} & 9x + 2y = 2 & - (1) \\ & 3x + 5y = 5 & - (2) \end{array}$$

Solⁿ: - Since the ratio of co-efficient of $y = 2/5$ and ratio of constants $= 2/5$ is equal

Therefore x can be assumed to be zero and we can directly substitute $x = 0$ in (1).

$$\therefore 2y = 2$$

$$\therefore y = 1$$

$$x = 0, y = 1$$

$$1) \quad 37x + 29y = 95 \quad - (1)$$

$$29x + 37y = 103 \quad - (2)$$

∴ Since the ratio of coefficients of x and y are interchanged
 ∴ We will add the eq (1) and eq (2) and subtract them later.

Add:-

$$\begin{aligned} \therefore 37x + 29y &= 95 \\ (+) 29x + 37y &= 103 \\ \hline 66x + 66y &= 198 \\ - (\div 3) \end{aligned}$$

Subtract:-

$$\begin{aligned} 37x + 29y &= 95 \\ (-) 29x + 37y &= 103 \\ \hline 8x - 8y &= -8 \\ - (\div 8) \end{aligned}$$

Divide eq (3) by 66

Divide eq (4) by 8

$$x + y = 3 \quad - (5)$$

$$x - y = -1 \quad (6)$$

Adding (5) and (6) to get x

$$\begin{aligned} x + y &= 3 \\ x - y &= -1 \\ \hline 2x &= 2 \end{aligned}$$

$$\therefore \boxed{x = 1}$$

Substitute $x = 1$ in (5)

$$\begin{aligned} 1 + y &= 3 \\ \hline y &= 2 \end{aligned}$$

$$\therefore x = 1, y = 2$$

$$\begin{aligned} \text{j)} \quad 12x + 17y &= 53 & - (1) \\ 17x + 12y &= 63 & - (2) \end{aligned}$$

solⁿ: - Since the ratio of coefficients of x and y are interchanged therefore we can add and subtract the equations.

Adding:-

$$\begin{aligned} 12x + 17y &= 53 \\ 17x + 12y &= 63 \\ \hline 29x + 29y &= 116 \\ &- (3) \end{aligned}$$

Divide eq (3) by 29

$$\begin{aligned} x + y &= 4 \\ &- (5) \end{aligned}$$

Subtracting:-

$$\begin{aligned} 12x + 17y &= 53 \\ (-) 17x + 12y &= 63 \\ \hline (-) \quad (-) \quad (-) \\ -5x + 5y &= -10 \\ &- (4) \end{aligned}$$

Divide eq (4) by 5

$$\begin{aligned} -x + y &= -2 \\ &- (6) \end{aligned}$$

Adding (5) and (6) to get y

$$\begin{aligned} x + y &= 4 \\ -x + y &= -2 \\ \hline 2y &= 2 \end{aligned} \quad \therefore y = \frac{2}{2} = \boxed{1}$$

Substituting $y = 1$ in (5)

$$x + 1 = 4$$

$$\therefore \boxed{x = 3}$$

$$x = 3, y = 1$$